

Is the contracted dim J
sharded in A or B?

No. Do two
non-contracted dims
share a mesh axis?

no

yes

Case 1
local
matmul,
no comms

Case 4
AllGather
one input
first

Yes. Is J sharded
the same way in
both inputs?

no

yes

Case 2
AllGather
the sharded
input

Case 3
partial sums,
then AllReduce
or Reduce-
Scatter

Case 4 can co-occur with 2 or 3: fix it first